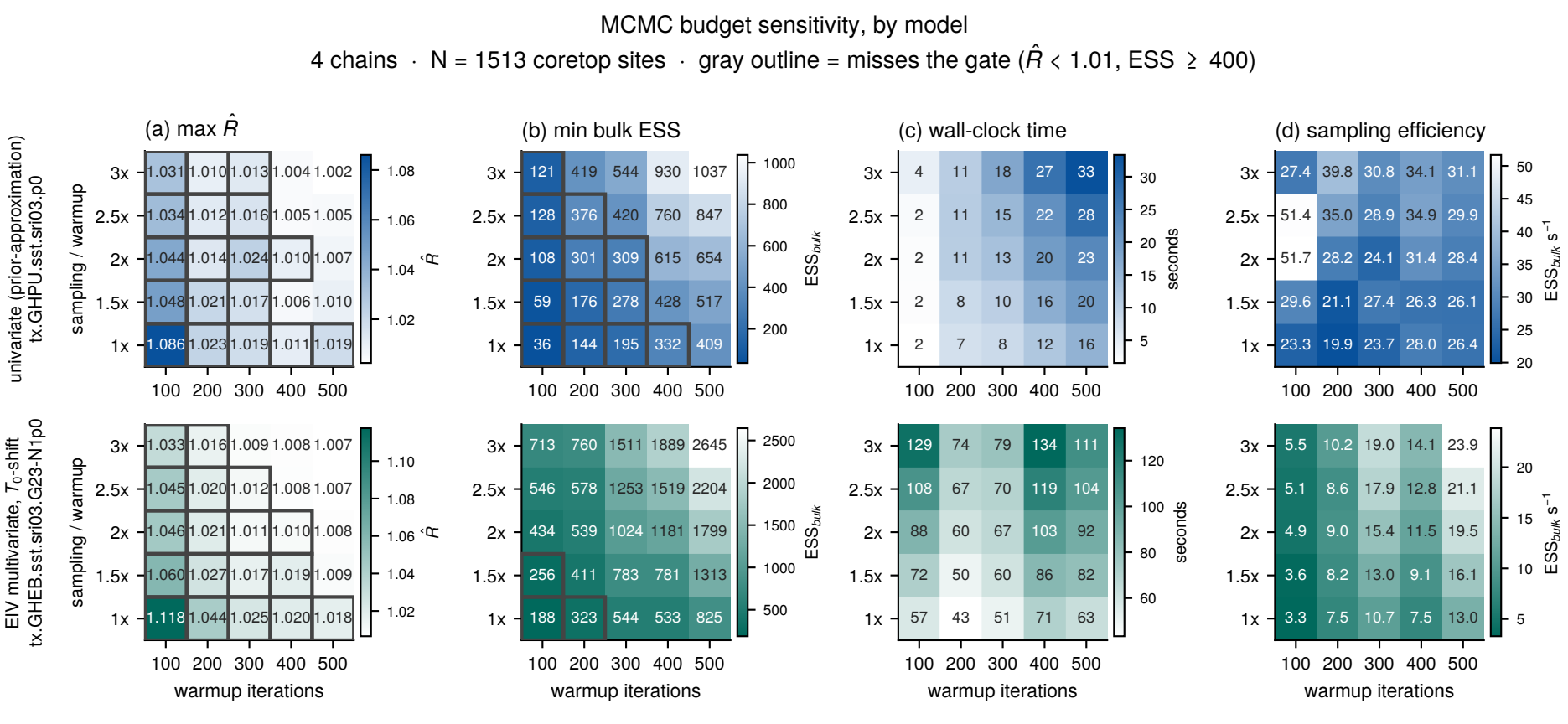
MCMC budget sensitivity, by model
4 chains · N = 1513 coretop sites · gray outline = misses the gate ($\hat{R}$ < 1.01, ESS ≥ 400)
(a) max $\hat{R}$
(b) min bulk ESS
(c) wall-clock time
(d) sampling efficiency
univariate (prior-approximation)
tx.GHPU.sst.sri03.p0
EIV multivariate, T_0-shift
tx.GHEB.sst.sri03.G23-N1p0
sampling / warmup
warmup iterations
$\hat{R}$
ESS_{bulk}
seconds
$ESS_{bulk}\ s^{-1}$
3x
2.5x
2x
1.5x
1x
100 200 300 400 500
1.031 1.010 1.013 1.004 1.002
1.034 1.012 1.016 1.005 1.005
1.044 1.014 1.024 1.010 1.007
1.048 1.021 1.017 1.006 1.010
1.086 1.023 1.019 1.011 1.019
121 419 544 930 1037
128 376 420 760 847
108 301 309 615 654
59 176 278 428 517
36 144 195 332 409
4 11 18 27 33
2 11 15 22 28
2 11 13 20 23
2 8 10 16 20
2 7 8 12 16
27.4 39.8 30.8 34.1 31.1
51.4 35.0 28.9 34.9 29.9
51.7 28.2 24.1 31.4 28.4
29.6 21.1 27.4 26.3 26.1
23.3 19.9 23.7 28.0 26.4
1.033 1.016 1.009 1.008 1.007
1.045 1.020 1.012 1.008 1.007
1.046 1.021 1.011 1.010 1.008
1.060 1.027 1.017 1.019 1.009
1.118 1.044 1.025 1.020 1.018
713 760 1511 1889 2645
546 578 1253 1519 2204
434 539 1024 1181 1799
256 411 783 781 1313
188 323 544 533 825
129 74 79 134 111
108 67 70 119 104
88 60 67 103 92
72 50 60 86 82
57 43 51 71 63
5.5 10.2 19.0 14.1 23.9
5.1 8.6 17.9 12.8 21.1
4.9 9.0 15.4 11.5 19.5
3.6 8.2 13.0 9.1 16.1
3.3 7.5 10.7 7.5 13.0